

CSDS 302 Discrete Math | Fall 2026 | Homework 6

Instructions

Section 5.6: 4, 12, 38

Correction: In question 25 of chapter 6.1, index i starts from 1, instead of 0.

Section 6.1: 1bd, 4, 12abgi (2 points each part), 25acdf(2 points each part), 29

Section 6.2: 17, 18, 25, 42

Section 9.2: 11c, 17ce, 21, 31de, 39bd

Section 9.3: 2b, 7cd, 17, 23bc

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Feedback

Score: 166/168 (98.81%)

Section 9.2

21c: -2

Homework 6 - Zelan Espanto

5.6: 4, 12, 38

#4 $d_k = k(d_{k-1})^2, k \geq 1$

$d_0 = 3$

$d_3 = 3(162)^2 = 78732$

$d_1 = 1(3)^2 = 9$

$d_4 = 4(78732)^2 = 24794911296$

$d_2 = 2(9)^2 = 162$

#12 $S_n = \frac{(-1)^n}{n!}, n \geq 0$

$S_k = \frac{-S_{k-1}}{k}, k \geq 1$

$S_0 = 1$

$S_1 = -1/1 = -1$

the series satisfies

$S_1 = -1/1 = -1$

$S_2 = -1/2$

since both sides are

$S_2 = -1/2 = -1/2$

$S_3 = -1/6 = -1/6$

the same

$S_3 = -1/6$

#38 (a) annual paying $\rightarrow 3\%$ or 0.03

every month: $\frac{0.03}{12} = 0.0025$ /mo.

$S_n = S_{n-1}(1.0025)$ monthly $S_n = S_{n-1}(1.03)$ /yr

(b) $S_0 = 10,000; S_1 = 10,025; S_2 = 10,050.08 \dots$

$S_{12} = 10304.16$

(c) APY = $\frac{10,304.16 - 10,000}{10,000} = 3.04\%$

6.1: 1bd, 12abgi, 25acdf, 29

(1)(b) $A = \{3; \sqrt{5-4} = 24 \text{ mod } 7\} = \{3; 3; 3\} = \{3\}$

$B = \{8 \text{ mod } 5\} = \{3\}$

$A \subseteq B$ and $B \subseteq A \Rightarrow A = B$

(d) $A = \{a, b, c\}$ $B = \{a, b, c\}$

$A \not\subseteq B$ and $B \not\subseteq A$

(12) R and rul $A^c = \{x < -3; x > 0\}$

number line

$A = \{x \in \mathbb{R} \mid -3 \leq x \leq 0\}$ $B = \{x \in \mathbb{R} \mid -1 < x < 2\}$

$C = \{x \in \mathbb{R} \mid 6 < x \leq 8\}$

(a) $A \cup B = \{-3; -2; -1; 0; 1\}$

(b) $A \cap B = \emptyset$

(c) $A^c \cap B^c = \{x \in \mathbb{R} \mid -\infty < x < -3, 2 \leq x < \infty\}$

(i) $(A \cap B)^c = \{x \in \mathbb{R} \mid -\infty < x \leq -1; 0 < x < \infty\}$

25acdf

29

(25) $R_i = \{x \in \mathbb{R} \mid 1 \leq x \leq 1 + \frac{1}{i}\} = [1; 1 + \frac{1}{i}]$

(a) $\bigcup_{i=1}^4 R_i = \{1; 2; \frac{3}{2}; \frac{4}{3}; \frac{5}{4}\}$

$R_1 = \{1, 2\}$

$R_2 = \{1; \frac{3}{2}\}$

$R_3 = \{1; \frac{4}{3}\}$

$R_4 = \{1; \frac{5}{4}\}$

(c) $R_1, R_2, R_3 \dots$ mutually disjoint.

No since every R_i contains a mutual element 1

(d) $\bigcup_{i=1}^{\infty} R_i = [1, \infty]$

(f) $\bigcup_{i=1}^{\infty} R_i = [1; 1 + \frac{1}{i}]$

(29) $\mathbb{R}$ all real numbers is $\{R^+, R^-, \{0\}\}$

partition of $\mathbb{R}$? $R^+ \cap R^- \cap \{0\} = \emptyset$ and

$R^+ \cap R^- \cap \{0\} = R$

$\therefore$ Disjoined

6.2: 17, 18, 25, 42

(17) sets A, B, C

$A \subseteq B \Rightarrow A \cup C \subseteq B \cup C$

Let $a \in A \cup C \Rightarrow a \in A$ or $a \in C$ by Un. Det

Case $a \in A$ and $A \subseteq B \Rightarrow a \in B \cup C$ by In Det

Case $a \in C \Rightarrow a \in B \cup C$

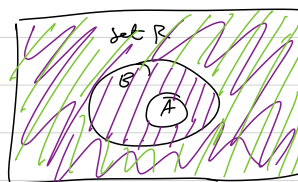
$a \in B \cup C \Rightarrow A \cup C \subseteq B \cup C$ TRUE

(18) A and B are sets. If $A \subseteq B$ then $B^c \subseteq A^c$

Let $a \in B^c; a \notin B$; since $A \subseteq B$

$a \notin A$, but $a \in A^c$ by Compl. of

$B^c \subseteq A^c \Rightarrow A \subseteq B \Rightarrow B^c \subseteq A^c$



$|A^c| > |B^c|$
or A^c is bigger than B^c

(25) mistake 1: if $x \in A \Rightarrow x \in (A - B)$ is not true

mistake 2: should have done show that

$x \in [(A - B) \cup (A \cap B)]$ & $x \in A$

$$(42) \bigcap_{i=1}^n (A_i - B) = (\bigcap_{i=1}^n A_i) - B$$

$$\bigcap_{i=1}^n = (A_0 - B \cap A_1 - B \cap A_2 - B \dots \cap A_n - B)$$

By distributive property
 $(A_0 \cap A_1 \cap A_2 \dots \cap A_n) - B$

Finite $\cap$ of integer therefore ^{this can be} expressed

$$\text{as } \bigcap_{i=1}^n (A_i - B) = (\bigcap_{i=1}^n A_i) - B$$

9.2: 11c, 17ce, 21, 31de, 39bd

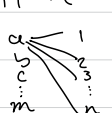
(11c) 2^6 Because if first 2 last one one it leaves with
 $2^{8-2} = 2^6 = 64$

(17) $\subseteq$ from 1000 to 9999 distinct digits $9 \cdot 8 \cdot 7 \cdot 6 = 3024$
 $\subseteq 9 \cdot 8 \cdot 7 \cdot 6 = 3024$

$$9999 - 1000 + 1 = 9000 \text{ total no}$$

Probability for distinct = $\frac{3024}{9000} = 33\%$

Odd: $\frac{50}{2} = 25\%$

(21) $|A| = m$ $|B| = n$

 2^n but for each in $m \cdot 2^{mn}$

$\subseteq$ B is the answer, A is toward n^m

$$\subseteq \frac{n^m}{2^{mn}} = \frac{n^m}{2^n 2^m} = \left(\frac{n}{2}\right)^m \cdot 2^n$$

(31) $\frac{d}{p_1^{a_1} p_2^{a_2} \dots p_m^{a_m}}$ and a_1, a_2, a_3
 $p_1^{a_1} p_2^{a_2} \dots p_m^{a_m} \Rightarrow p_i^{a_i}$ has $(a_i + 1)$ distinct div
because p is multiplied a_i times plus 1 as divisor
 $p_2^{a_2} \Rightarrow a_2 + 1$ so ...
 $p_1^{a_1} p_2^{a_2} \dots p_m^{a_m} \Rightarrow (a_1 + 1)(a_2 + 1) \dots (a_m + 1)$

$$\subseteq 12 \text{ div } 12 = 3 \cdot 2 \cdot 2 = 6 \cdot 2 = 4 \cdot 2$$

Using Part (d)

4) $(a_1 + 1) = 3 \Rightarrow a_1 = 2$ 2) $(a_1 + 1) = 6 \Rightarrow 5$
 $(a_2 + 1) = 2 \Rightarrow a_2 = 1$ 3) $(a_2 + 1) = 2 \Rightarrow 1$
 $(a_3 + 1) = 2 \Rightarrow a_3 = 1$ 3) $(a_1 + 1) = 4 \Rightarrow a_1 = 3$
 $(a_2 + 1) = 2 \Rightarrow a_2 = 1$

For (1) $2^2 \times 5 \times 3 = 60$

For (2)

$2^5 \cdot 3 = 96$

For (3)

$2^3 \cdot 3 = 12$

(39) $\subseteq$ all are UNIQUE
 $9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = 60480$
 $\frac{d}{7 \cdot 6 \cdot 5 \cdot 4 = 840}$

9.3: 2b, 7cd, 17, 23bc

(2) $\subseteq 2:16^1$ $3:16^3$ $4:16^4$ $5:16^5$ $16^2 + 16^3 + 16^4 + 16^5 = 1118464$

(7) 3-5 symbols 26 uppercase 0-9 14 symbols

$\subseteq$ total # of symbols $26 + 10 + 14 = 50$

all the pass $50^3 \neq 50^4 + 50^5 = 318875000$

Imagine pass $50 \cdot 49 \cdot 48 + 50 \cdot 49 \cdot 48 \cdot 47 + 50 \cdot 49 \cdot 48 \cdot 47 \cdot 46$
 259896000 all unique = 589790000

$\frac{d}{589790000} = 19\%$

(17) $\subseteq 16 \cdot 15 \cdot 14 \cdot 13 = 43680$
 $\subseteq 16 \cdot 16 \cdot 16 \cdot 16 = 65536$
 $65536 - 43680 = 21856$
 $\subseteq \frac{21856}{65536} = 33.3\%$

(23) $\subseteq \frac{357}{1000} \cdot 100\% = 35.7\%$
 $\subseteq \frac{1000}{2} = 500$
 $\frac{1000}{9} = 111$
 $\frac{1000}{18} = 55$

$500 + 111 - 55 = 556$ mod of 2 or 9